Lily's Florist: Managing AI Engineering Project

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October 25, 2026

Note from the author: This document combines my own writing with AI-generated text from ChatGPT Codex and Figma Make. See Appendix C and the References for supporting material.

Introduction

This project outlines a proposed website for Lily's Florist to simplify online flower ordering, support inventory and order management, and provide AI customer assistance. The following sections bring together the interview synopsis, system requirements, user stories, and a user journey. The full interview, project checklist, and wireframe design conversation appear as appendices.

Synopsis

Key Topics and Responses

- **Pain points:** Manual order entry and stock tracking cause missed details, duplicate orders, and overselling; repetitive inquiries interrupt staff during busy periods.
- **Business goals:** Simplify online ordering, reduce administrative work, and grow the business while preserving personal customer service.
- **Sales and service targets:** Increase online sales by 15–20% in year one and 30–40% within three years, achieve 90% customer satisfaction, and automate 70% of routine inquiries.
- **Efficiency targets:** Cut order-processing time by at least 50% and flower waste by 25%, while growing normal daily orders from 20–30 to 40–50.

Categories of User Requirements

- **User groups:** Customers and visitors need shopping and support tools; florists, delivery staff, and Lily need role-appropriate operational access, while event customers need quotes.
- **Functional requirements:** Support browsing, ordering, payments, scheduling, inventory, notifications, staff workflows, and AI assistance.
- **Nonfunctional requirements:** Provide secure, reliable, mobile-friendly service with screen reader and keyboard support, readable interfaces, and English and Spanish at launch.
- **Privacy and AI boundaries:** Verify identity before exposing order details and restrict staff access by role; AI must not independently change orders or issue refunds.

Website Features

- **Shopping and payment:** Offer guest checkout, arrangement options, extras, and basic discount codes, with cards and digital wallets online or cash/card for pickup.
- **Fulfillment:** Offer delivery within 15 miles, capacity-controlled slots, and bookings up to 30 days ahead, with automatic email updates, optional SMS, and secure guest tracking links.
- **Inventory and exceptions:** Reserve stock for 15 minutes during checkout, track stock through preparation, and support payment retries, approved substitutions, cancellations, and refunds.
- **AI support:** Answer routine questions around the clock using maintained shop information and live stock/order data, with staff handoff or next-business-day follow-up when needed.
- **Content and administration:** Include Home, Shop, Gallery, About Us, Contact, Delivery & FAQs, and Order Tracking, plus staff tools for orders, inventory, deliveries, and content updates.

Constraints

- **Schedule:** Lily prefers a January 15, 2027 launch, allowing testing and staff training before Valentine's Day.
- **Budget:** Initial spending is $12,000–$18,000, with approximately $300–$500 monthly for hosting, maintenance, and AI services.
- **Capacity and continuity:** Support 150–200 daily orders during peaks with simple workflows for a small team and minimal disruption to ongoing business.
- **Existing systems:** Retain Square and Gmail, connect online payments to Square if possible, and preserve existing customer records and product information during migration.

Other Relevant Information

- **Release priorities:** Launch ordering, scheduling, inventory, staff tools, and basic AI support first; defer accounts, loyalty automation, event booking, blogging, and forecasting to Version 2.
- **Content ownership:** Lily maintains prices, policies, promotions, and FAQs, while florists maintain product details and availability; listings are reviewed weekly and seasonally refreshed.
- **Clarifications needed:** Confirm how unpaid pickup orders enter the staff workflow and reserve stock, whether AI recommendations belong at launch or in Version 2, and whether the stated flower budget includes delivery and tax.

Source: Full interview transcript.

Requirements

These draft requirements preserve the details extracted from the original user stories. Each requirement identifies its source story. Functional requirements describe behavior; nonfunctional requirements describe interface, security, or privacy constraints. This is a selected set of requirements for the project, not a complete launch specification; the synopsis records Lily's broader goals. Detailed catalog categories, sorting options, and staff pricing fields are proposed design choices requiring client confirmation.

The AI flow below uses public catalog information and shopping-session context only. Lily's broader request for verified order support and contact collection requires a separate privacy design; NFR-04 applies to the selected flow.

Functional Requirements

FT-01: Inventory Browsing

ID	Requirement	Source
FR-01	The website shall provide a Shop navigation control that opens the shop's inventory when selected.	US-01
FR-02	Selecting the category control shall display an inventory category menu using touch, mouse, or keyboard input.	US-02
FR-03	The category menu shall include individual flowers, arrangements (identical), arrangements (mixed), clearance, seasonal, flower health supplements, containers and vases, and accessory items.	US-02
FR-04	Selecting a category shall display its items and allow keyword searches within that category.	US-03
FR-05	Category listings shall mark out-of-stock items as unavailable and prevent adding them to the cart.	US-03
FR-06	Category listings shall include items on sale.	US-03
FR-07	Customers shall be able to sort category listings by price, season, color(s), holiday, and in-stock status.	US-04
FR-08	Customers shall be able to filter category listings by price, season, color(s), holiday, and in-stock status.	US-04

FT-02: Shopping Cart and Purchasing

ID	Requirement	Source
FR-09	Selecting an item's plus (+) control shall add that item to the customer's shopping cart.	US-05
FR-10	After an item is added, the website shall display a cart drawer containing the selected items, using a bottom sheet on mobile.	US-06
FR-11	The cart drawer shall provide a Check Out button.	US-07
FR-12	Selecting Check Out shall open the checkout page with the customer's cart contents.	US-07

ID	Requirement	Source
FR-13	Customers shall be able to change item quantities on the checkout page.	US-08
FR-14	Customers shall be able to remove items from their cart on the checkout page.	US-08
FR-15	Selecting Confirm on the checkout page shall prompt the customer to select an available payment method.	US-09
FR-16	Pickup orders shall offer cash as a payment option.	US-09
FR-17	Online payment options shall include credit and debit card payments.	US-09
FR-18	Online payment options shall include Apple Pay and Google Pay, subject to payment-provider support.	US-09
FR-19	After payment is completed, the website shall provide an invoice and receipt.	US-10
FR-20	The invoice and receipt shall include the order number.	US-10
FR-21	After payment is completed, the system shall email the invoice and receipt, including the order number, to the customer.	US-10

FT-03: AI Assistance

ID	Requirement	Source
FR-22	Selecting the AI chat icon shall open a conversation with an AI agent.	US-11
FR-23	The AI agent shall have access to publicly available website information for answering customer questions.	US-12
FR-24	The AI agent's shop information shall include store history, hours, location, contact information, and current public product availability.	US-12
FR-25	Subject to the privacy constraints below, the AI agent shall have access to the customer's browsing history within the current website session.	US-13
FR-26	Subject to the privacy constraints below, the AI agent shall have access to the customer's current shopping cart contents.	US-13
FR-27	The AI agent shall use available session and cart context to answer questions about the customer's current shopping selections.	US-13

FT-04: Staff Web Portal

ID	Requirement	Source
FR-28	Associates shall be able to log into the web portal and view the full shop inventory.	US-15
FR-29	The inventory view shall display the quantity of each item.	US-15
FR-30	The inventory view shall display each item's cost of goods.	US-15
FR-31	The inventory view shall display each item's MSRP.	US-15

ID	Requirement	Source
FR-32	The inventory view shall display each item's current display price.	US-15
FR-33	Logged-in associates shall be able to view all pending orders.	US-16
FR-34	Each pending order shall display its invoice number.	US-16
FR-35	Each pending order shall display its sale price.	US-16
FR-36	Each pending order shall display the ordered items and their quantities.	US-16
FR-37	Logged-in administrators shall be able to add web portal users, including associates.	US-17
FR-38	Logged-in administrators shall be able to delete web portal users, including associates.	US-17
FR-39	Logged-in administrators shall be able to issue refunds for transactions.	US-18
FR-40	Logged-in administrators shall be able to modify transaction sale prices.	US-18

Nonfunctional Requirements

ID	Requirement	Source
NFR-01	The inventory navigation control shall be located at the top of the website.	US-01
NFR-02	Each item shall have a plus (+) icon as its add-to-cart control.	US-05
NFR-03	The AI chat icon shall appear in the bottom-right corner of the web interface while the customer uses the website.	US-11
NFR-04	The AI agent shall not have access to customers' credit card numbers, names, phone numbers, or email addresses.	US-14
NFR-05	The website shall not track customers' personal identifying information without their consent.	US-14

User Stories

Project Epic

We want to create a simple online flower-ordering experience for Lily's Florist, where customers can browse inventory, save their selection, pay for their order, track current orders, and receive AI assistance through a responsive interface. Employees should be able to track orders and inventory, schedule deliveries, and access tools appropriate to their roles; customer account histories are planned for Version 2.

Project Features and User Stories

Note from the author: The feature and story format is adapted from practices I use at work and learned in school. Each story identifies a role, need, and intended outcome.

FT-01: Browse and Search Inventory

Customers need to browse the catalog and find items using categories, search, sorting, and filters.

- **US-01:** As a Lily's Florist Customer, I want navigation at the top of the website, so that I can navigate to the shop's inventory.
- **US-02:** As a Lily's Florist Customer, I want to browse inventory categories, so that I can find the types of items I am looking for.
- **US-03:** As a Lily's Florist Customer, I want to browse and search items in an inventory category, so that I can explore the shop's selection and availability.
- **US-04:** As a Lily's Florist Customer, I want to sort and filter inventory, so that I can find items that match my preferences.

FT-02: Purchase Items

Customers need a clear checkout flow with secure payment and a record of their purchase.

- **US-05:** As a Lily's Florist Customer, I want to add items to my shopping cart, so that I can save my selection for purchase.
- **US-06:** As a Lily's Florist Customer, I want to see the items in my shopping cart, so that I can keep track of my selection.
- **US-07:** As a Lily's Florist Customer, I want to proceed to checkout from my shopping cart, so that I can purchase my selected items.
- **US-08:** As a Lily's Florist Customer, I want to modify my order at checkout, so that I can adjust my selection before purchasing.
- **US-09:** As a Lily's Florist Customer, I want to pay using a variety of available methods, so that I can complete my purchase using a method that works for me.
- **US-10:** As a Lily's Florist Customer, I want an invoice and receipt with an order number, so that I have a record of my purchase.

FT-03: Get AI Assistance

Customers need contextual shopping assistance while retaining control over their personal data.

- **US-11:** As a Lily's Florist Customer, I want to chat with an AI agent while using the website, so that I can get assistance when needed.
- **US-12:** As a Lily's Florist AI Agent, I want access to publicly available website and store information, so that I can answer customers' questions about the shop.
- **US-13:** As a Lily's Florist AI Agent, I want context about a customer's shopping experience, so that I can provide helpful information specific to their visit.
- **US-14:** As a Lily's Florist Customer, I want control over the use of my personal identifying information, so that I can receive AI assistance while protecting my privacy.

FT-04: Use Staff Tools

Associates and administrators need tools for inventory, orders, access, and transaction management.

- **US-15:** As a Lily's Florist Associate, I want to view the full shop inventory, so that I can check stock quantities and pricing information for my job.
- **US-16:** As a Lily's Florist Associate, I want to view pending orders, so that I can see what customers have ordered and support order preparation.
- **US-17:** As a Lily's Florist Administrator, I want to manage web portal users, so that I can maintain access for associates and other users.
- **US-18:** As a Lily's Florist Administrator, I want to modify transactions, so that I can manage refunds and sale price adjustments.

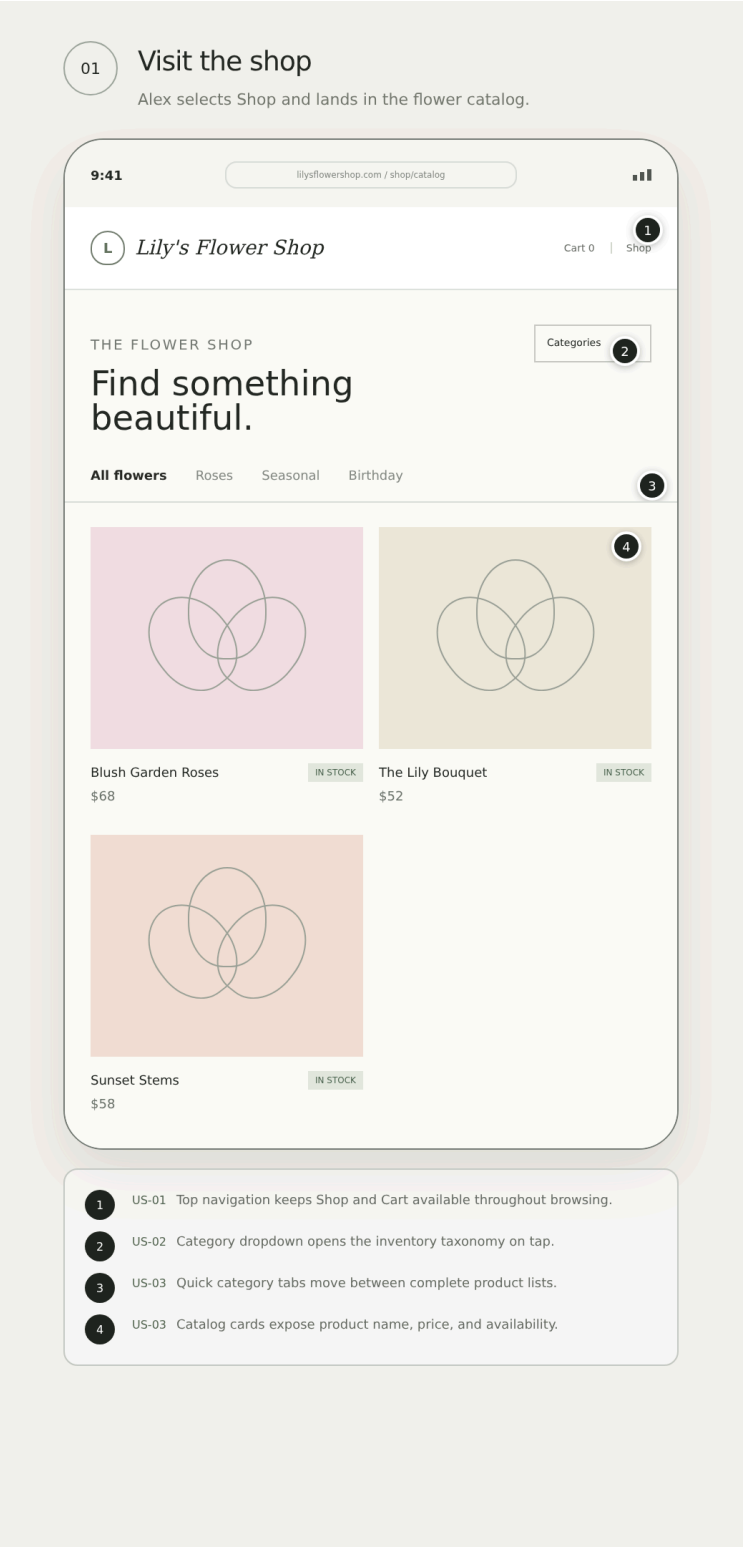
User Journey: Finding Flowers with AI Assistance

Alex is a first-time visitor looking for pink roses as a birthday gift. The goal is to find available flowers and proceed to checkout. The journey spans inventory browsing (FT-01), AI assistance (FT-03), and the cart-to-checkout handoff (FT-02). The wireframes use a mobile-first layout and the display name "Lily's Flower Shop" for the same hypothetical business.

Step	User action	System response	Related stories
1. Visit	Alex opens the website and selects Shop.	Displays the flower catalog.	US-01
2. Search	Alex searches for roses.	Displays matching flowers with prices and availability.	US-03
3. Ask AI	Unsure which colors are available, Alex opens chat and asks, "Are pink roses in stock?"	Checks current inventory, confirms availability, and links to the matching flowers.	US-11, US-12
4. Check out	Alex adds the pink roses to the cart and clicks "Check Out."	Displays the cart drawer and hands off to checkout. The mockup ends before payment.	US-05, US-06, US-07

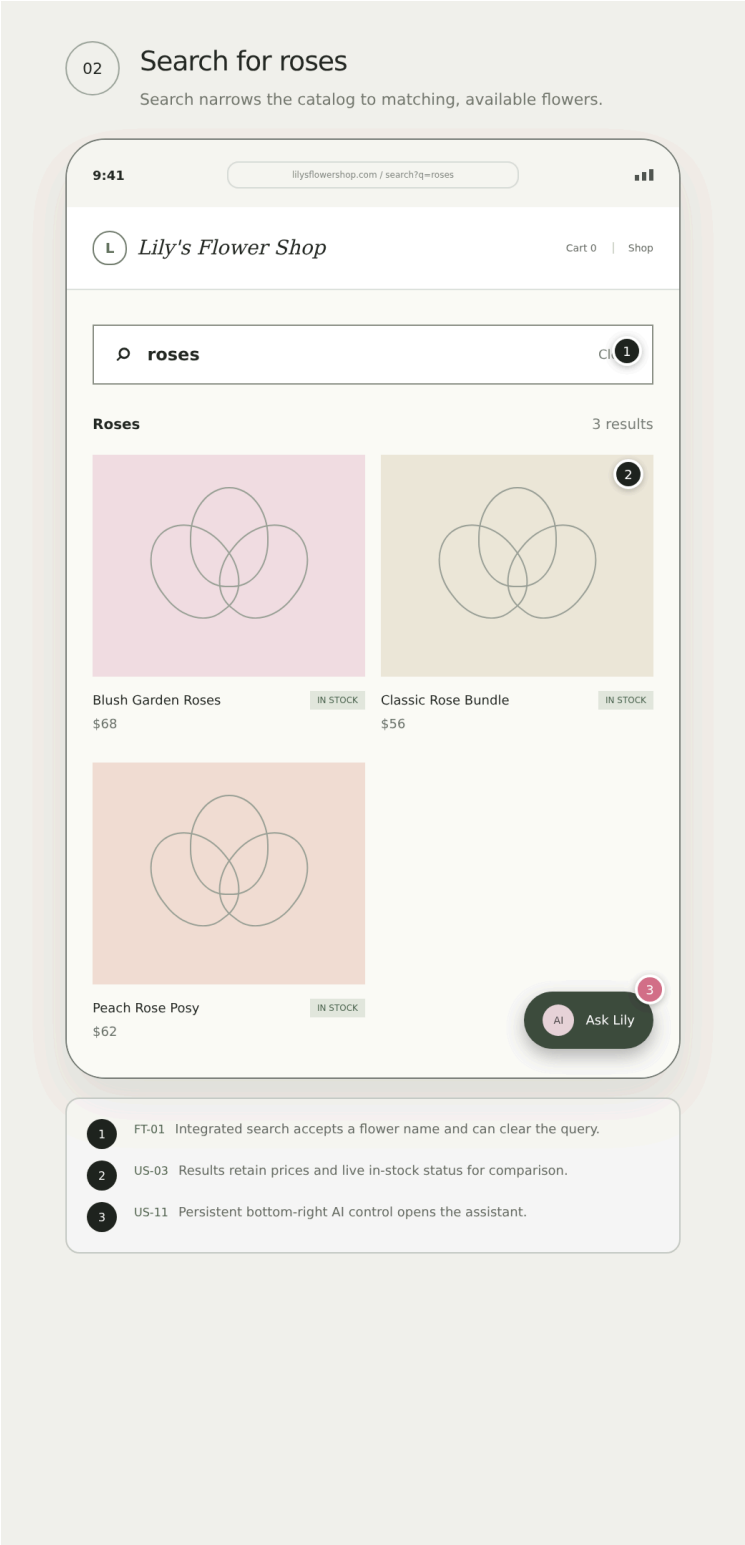
If pink roses are unavailable, the AI suggests an available color so Alex can choose an alternative and continue to checkout. If no alternative is suitable, Alex can return to browsing or leave without placing an order. The wireframes illustrate the flow; they do not demonstrate a live inventory or payment integration.

1. Visit the Shop



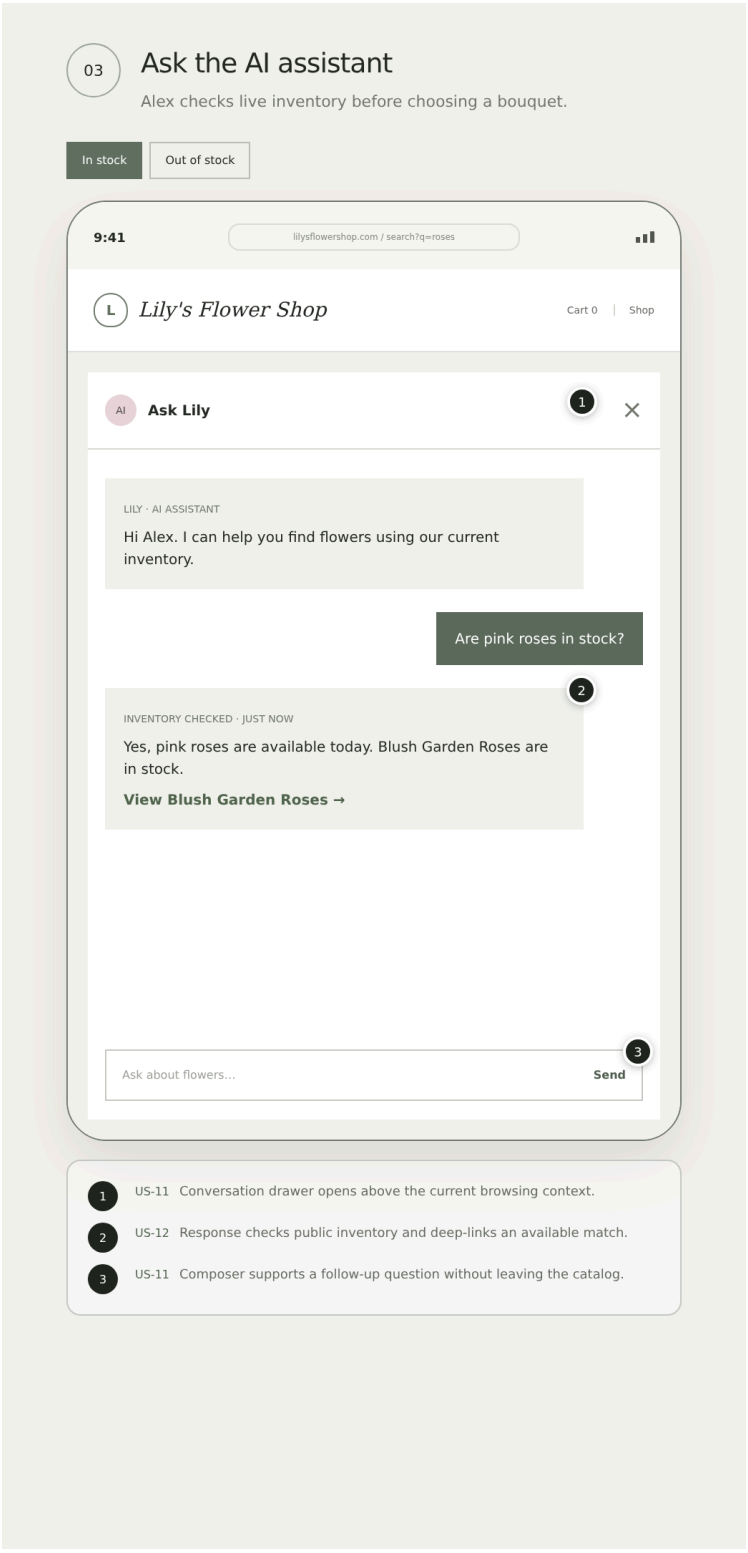
The top navigation and category control support US-01–US-03 and FR-01–FR-04; NFR-01 specifies the navigation placement.

2. Search for Roses

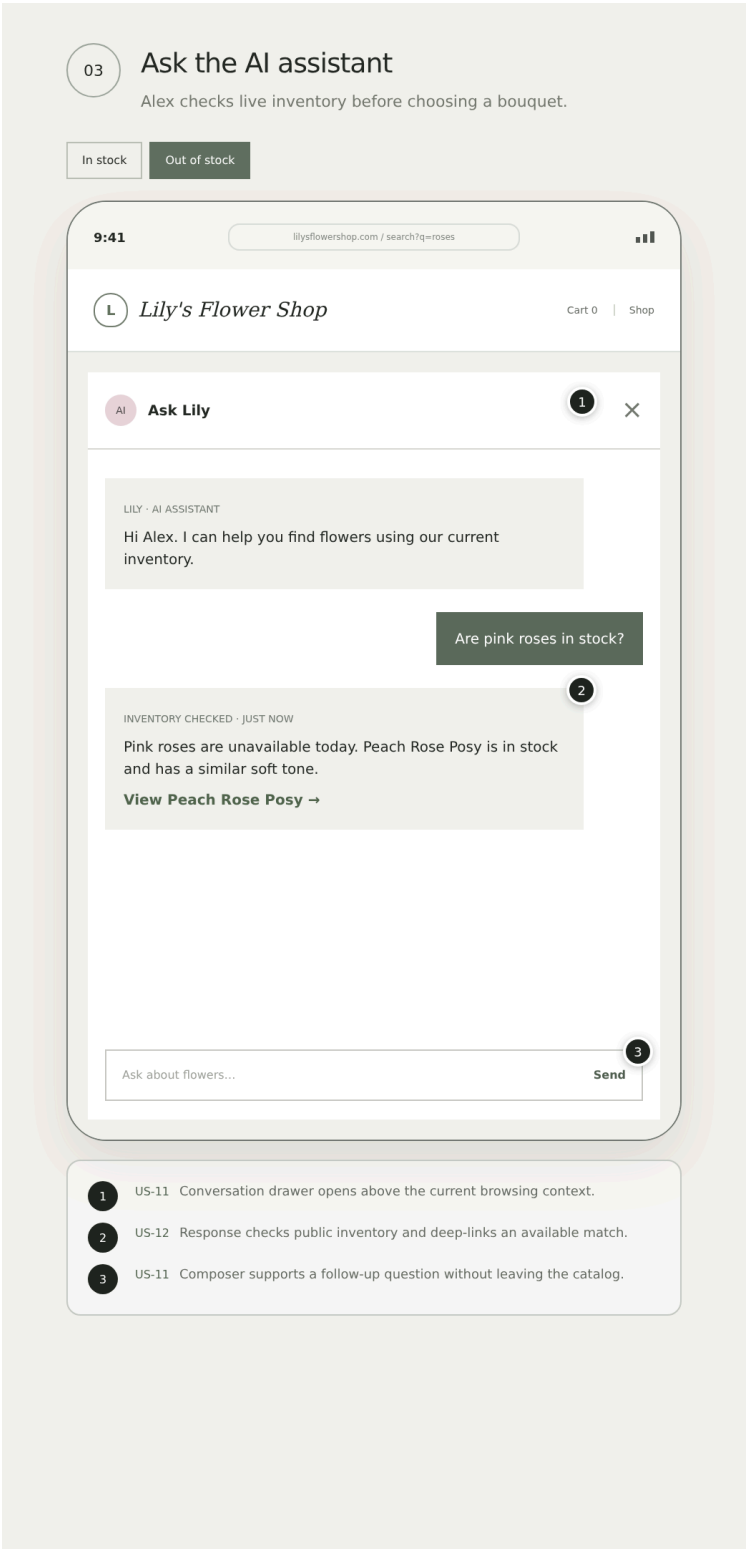


The search and results support US-03 and FR-04–FR-06; availability labels distinguish items that can be purchased from those that are out of stock.

3. Ask AI About Availability

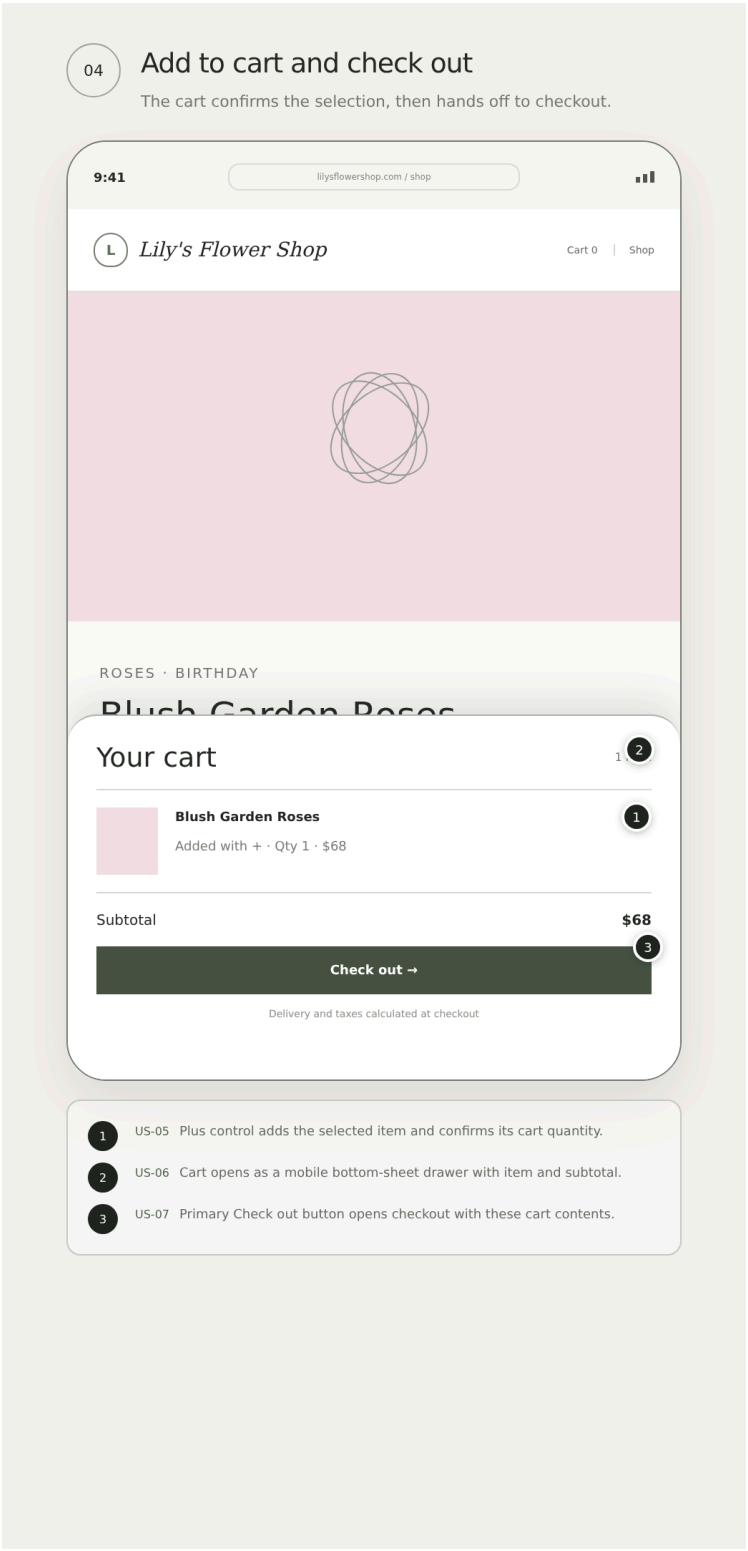


3b. AI Suggests an Available Alternative



The chat control and public availability response support US-11–US-12 and FR-22–FR-24. The alternate state offers an available color when pink roses are unavailable.

4. Proceed to Checkout



The plus control, cart drawer, and Check Out button support US-05–US-07, FR-09–FR-12, and NFR-02; payment is outside this illustrated journey.

Appendix A: Interview with Lily

Conversation

View the shared interview.

The transcript is preserved as recorded. Open questions in the synopsis include the example AI response that exceeds the customer's stated $60 budget once delivery is added.

1. Neil

Hi Lily, I'm Neil, a website developer, and I am going to help you with your new, state-of-the-art website, with AI integration and e-commerce integration. Can I ask you a few questions about your site?

2. Lily

Hi Neil! 🌸 It's lovely to meet you, and I'm delighted to have you helping me create a new website for Lily's Florist!

I've been running my flower shop for over 10 years, and I'm excited about bringing more of my business online.

I'm particularly interested in making ordering flowers easier for my customers, managing my growing number of orders, and using AI to improve customer service.

I'm happy to answer your questions and share my ideas about what I'd love the website to do! 🌷

3. Neil

What are your short-term and long-term goals for the website?

4. Lily

Hi Neil! 🌸 I'm glad you asked! I have some clear goals for Lily's Florist.

Short-term goals (first 6–12 months):

- Launch an attractive, easy-to-use website with online ordering and secure payments.
- Allow customers to schedule deliveries and pickups.
- Manage orders and flower inventory more efficiently.
- Introduce AI customer support to answer common questions 24/7.
- Showcase my floral arrangements through a beautiful gallery.

Long-term goals (1–3 years):

- Increase online sales by 30–40% and attract new customers.
- Introduce customer accounts, loyalty rewards, and personalized recommendations.
- Expand delivery services and grow my wedding and corporate event business.
- Use sales trends to forecast demand and reduce flower waste.
- Build a stronger online presence through seasonal promotions, social media, and a flower care blog.

Ultimately, I want the website to help my business grow while maintaining the personal touch my customers love! 🌷

5. Neil

How will you judge whether the site is successful, and what measurable improvements would you like to see compared with today?

6. Lily

Hi Neil! 🌸 For me, success means more sales, happier customers, and less time spent on paperwork!

I'd like to measure these improvements against today's performance:

- Online sales: Increase sales by 15–20% in the first year and 30–40% within three years.
- Order processing: Reduce the time spent handling each order by at least 50%.
- Customer satisfaction: Achieve a 90% satisfaction rating and encourage repeat purchases.
- Inventory management: Reduce flower waste by 25% through better stock tracking.
- AI customer support: Automatically answer at least 70% of routine customer inquiries.
- Order capacity: Increase our normal 20–30 daily orders to 40–50, while comfortably handling 100+ during holidays.

Most importantly, I'll consider the website successful if it helps Lily's Florist grow without losing the personal, caring service my customers love! 🌷

7. Neil

Which parts of handling orders, managing inventory, and answering customer questions cause the most frustration today? Can you describe a recent example?

8. Lily

Hi Neil! 🌸 My biggest frustrations are the amount of manual work and the difficulty keeping everything organized, especially during busy periods!

- Order management: I currently handle orders through phone calls, in-store visits, and a basic online form. Manually recording everything takes time and sometimes leads to missed details or duplicate orders.
- Inventory management: I track flower stock manually, making it difficult to know exactly what's available. Sometimes I accept orders without realizing certain flowers are running low.
- Customer inquiries: My staff and I repeatedly answer questions about prices, flower availability, delivery times, and opening hours, taking us away from arranging flowers.

A recent example: Last Mother's Day, we received over 100 orders in one day. We accidentally accepted several orders for pink peonies after running out of stock. I had to contact customers, suggest alternatives, and reorganize deliveries.

I'd love a website that automatically updates inventory, keeps orders organized, and uses AI to answer routine questions, allowing me to spend more time creating beautiful arrangements! 🌷

9. Neil

Who will use the site, including customers and staff, and what does each group need to accomplish?

10. Lily

Hi Neil! 🌸 I expect several groups of people to use Lily's Florist's website, each with different needs.

- Customers: Browse arrangements, place orders, pay securely, schedule deliveries or pickups, track orders, and earn loyalty rewards.

- Florists and shop staff: View incoming orders, check special requests, update order statuses, and manage daily flower inventory.
- Delivery staff: View delivery addresses, schedules, special instructions, and mark deliveries as completed.
- Me (shop owner): Manage products, prices, inventory, promotions, staff access, and review sales reports.
- Event customers: Request quotes for weddings, corporate events, and other special occasions.
- Website visitors: Explore our gallery, read flower care tips, check opening hours, and receive assistance through AI customer support.

My priority is making the website simple for customers while giving my staff the tools to manage busy days efficiently. I'd also like each staff member to access only the information needed for their role! 🌷

11. Neil

Which capabilities are essential for launch, and which could wait for a later version?

12. Lily

Hi Neil! 🌸 I'd like to prioritize features that help us manage orders efficiently and give customers a smooth shopping experience.

Essential for launch (Version 1):

- Online ordering with secure payments and simple checkout.
- Delivery and pickup scheduling with order confirmations.
- Inventory tracking that automatically updates when orders are placed.
- Staff dashboard to manage orders, stock, and deliveries.
- AI customer support for common questions, with the option to contact staff.
- Mobile-friendly website with a floral gallery, contact details, and opening hours.

Features for later versions (Version 2):

- Customer accounts, order history, and loyalty rewards.
- Personalized AI flower recommendations.
- Wedding and corporate event booking tools.
- Blog, seasonal promotions, and social media integration.
- Advanced sales reports and inventory forecasting.

My main priority: I want Version 1 to prevent overselling flowers, reduce manual work, and make ordering easier. We can introduce more sophisticated features once the essentials are working reliably! 🌷

13. Neil

Could you walk me through an ideal customer visit, from arriving on the site to receiving an order, including what the customer should see at each step?

14. Lily

Hi Neil! 🌸 I'd love customers to have a simple, enjoyable shopping experience from start to finish!

- Homepage: Customers see beautiful floral arrangements, seasonal specials, and clear Shop Now options.
- Browsing: They explore bouquets by occasion, price, or flower type, with photos, descriptions, prices, and availability.

- Selecting: They choose an arrangement, size, quantity, and optional extras such as chocolates or a greeting card.
- Checkout: They enter recipient details, a personal message, and choose delivery or pickup with an available date and time.
- Payment: They review the total cost, pay securely, and receive an immediate order confirmation by email.
- Order preparation: Customers receive updates when our florists begin preparing their flowers.
- Delivery: They receive notification when their order is out for delivery and confirmation once it arrives.
- After delivery: Customers can leave feedback and receive flower care tips.

My biggest priority is ensuring customers always see accurate flower availability, clear pricing, and reliable delivery information. I'd also love AI assistance throughout their visit to help them choose the perfect flowers! 🌷

15. Neil

What products and services should customers be able to order online, and what choices or details are needed for custom arrangements or event orders?

16. Lily

Hi Neil! 🌸 I'd love customers to be able to order our full range of flowers and services online!

Products and services:

- Bouquets: Birthday, anniversary, sympathy, romantic, and seasonal arrangements.
- Floral gifts: Roses, mixed bouquets, potted plants, chocolates, balloons, and greeting cards.
- Special occasions: Wedding flowers, corporate arrangements, centerpieces, and event decorations.
- Subscriptions: Weekly or monthly flower deliveries for homes and businesses.
- Custom arrangements: Customers should specify their budget, preferred flowers, colors, occasion, arrangement size, delivery date, personal message, and any flowers to avoid. They should also be able to upload inspiration photos.
- Wedding and event orders: I'd need the event date, venue, estimated guest count, types and quantities of arrangements, color theme, budget, and setup requirements. Customers should be able to request a personalized quote rather than paying immediately.

My priority: Standard arrangements should be available for immediate online purchase, while custom and event orders should allow me to review requests, confirm flower availability, and provide accurate pricing before accepting payment. 🌷

17. Neil

Will customers pay directly on the site, pay in person, or have both options? Which payment methods should be available?

18. Lily

Hi Neil! 🌸 I'd love to offer both online and in-person payments so customers can choose what works best for them.

Payment options I'd like to offer:

- Credit/debit cards: Visa, Mastercard, American Express, and Discover.
- Digital wallets: Apple Pay and Google Pay for convenient checkout.
- In-store payments: Cash or card for customers collecting their orders.
- Event orders: Deposits online, with the remaining balance paid before the event.

My preferences:

- Online delivery orders should require full payment before preparation.
- Customers choosing pickup can pay online or in-store.
- Payments must be secure, with immediate email receipts and confirmations.
- Refunds and cancellations should be easy for me to manage.

My priority is making payments convenient, secure, and straightforward for both customers and staff! 🌷

19. Neil

If a payment fails or the customer leaves checkout before finishing, what should happen to the order, any reserved inventory, and the customer's progress?

20. Lily

Hi Neil! 🌸 I'd like failed payments and abandoned checkouts to be handled smoothly, without frustrating customers or causing inventory problems.

- Failed payments: The order should remain pending, not confirmed. Customers should see a friendly error message and be able to retry or choose another payment method.
- Reserved inventory: Flowers should be temporarily reserved for 15 minutes during checkout. If payment isn't completed, the stock should automatically become available again.
- Abandoned checkout: Customers should be able to return and continue where they left off, provided their selected flowers are still available.
- Saved progress: Their cart, delivery details, and personal messages should be preserved, but payment card information shouldn't be stored.
- Reminders: Customers who have provided their email and agreed to reminders could receive a gentle reminder about their unfinished order.
- Staff notifications: I'd like staff alerted if a payment is received but the order isn't confirmed, so we can resolve it quickly.

My priority is preventing accidental double payments, overselling flowers, and losing potential customers because of checkout difficulties! 🌷

21. Neil

What delivery and pickup options should customers have, including service areas, fees, cutoff times, and available dates or time slots?

22. Lily

Hi Neil! 🌸 I'd like to offer flexible delivery and pickup options while keeping scheduling manageable for my staff.

- Delivery areas: Local delivery within 15 miles of the shop, with address eligibility checked during checkout.
- Delivery fees: $8 within 5 miles, $12 for 5–10 miles, and $18 for 10–15 miles. Free delivery for orders over $100 within 10 miles.
- Delivery times: Monday–Saturday, with morning (9 AM–12 PM) and afternoon (12 PM–5 PM) slots.
- Cutoff times: Same-day delivery for orders placed before 11 AM, subject to flower availability and delivery capacity.
- Store pickup: Free pickup Monday–Saturday, 9 AM–6 PM, with at least two hours' preparation time.

- Future bookings: Customers should be able to schedule deliveries or pickups up to 30 days ahead.

My priority: The website should display only available dates and time slots, automatically prevent overbooking, and allow me to adjust delivery capacity during busy holidays like Valentine's Day and Mother's Day! 🌷

23. Neil

If a requested delivery time is unavailable, a delivery fails, or an order is delayed, what should the customer be told and what alternatives should they have?

24. Lily

Hi Neil! 🌸 I want customers to feel informed and reassured whenever delivery problems occur.

- Unavailable delivery time: The website should immediately explain that the slot is fully booked and suggest the next available time, another date, or free store pickup.
- Delayed orders: Customers should receive an email or SMS explaining the delay, the updated estimated delivery time, and how to contact us.
- Failed deliveries: If nobody is available or the address is incorrect, our driver should contact the customer and notify the shop.
- Alternative arrangements: Customers should be able to request redelivery, reschedule, or arrange store pickup. Any additional delivery fees must be explained beforehand.
- Refunds and cancellations: If we cannot fulfill an order as promised, I'd like to offer a suitable alternative or a full refund.

My priority: Staff should be alerted immediately when something goes wrong, and customers should never be left wondering where their flowers are. I'd also like to personally handle serious delivery complaints to maintain our reputation for excellent service! 🌷

25. Neil

How should customers request order changes, cancellations, or refunds, and what restrictions should apply once preparation or delivery has begun?

26. Lily

Hi Neil! 🌸 I'd like customers to have a simple way to request changes, cancellations, or refunds while protecting my shop from unnecessary flower waste.

- Requesting changes: Customers should use a link in their confirmation email or contact us by phone to update delivery details, messages, or arrangements.
- Before preparation: Changes and cancellations should be free if requested at least 24 hours before the scheduled delivery or pickup.
- Once preparation begins: Changes should require staff approval, and cancellations may receive only a partial refund depending on flowers and materials already used.
- During delivery: Customers cannot cancel or change the delivery address without contacting us. Redelivery fees may apply.
- Refund requests: Customers should submit requests through the website, ideally within 48 hours of delivery, with photos if flowers arrive damaged.
- Staff approval: I'd like to personally review refunds and exceptions, with customers notified by email of the decision.

My priority: Customers should clearly understand our policies before checkout, while I retain flexibility to resolve genuine problems fairly. If an error is ours, I'd offer a replacement or full refund! 🌷

27. Neil

Could you walk me through how staff should receive, prepare, and complete an order, including how inventory should change along the way?

28. Lily

Hi Neil! 🌸 I'd love a clear, step-by-step process that keeps my team organized and prevents inventory mistakes.

- Order received: Staff get an alert when payment is confirmed. The order appears on our dashboard with customer details, flowers, special requests, and delivery information.
- Inventory reserved: Required flowers and supplies are automatically reserved when the order is confirmed, reducing available stock so we don't oversell.
- Order preparation: A florist marks the order as In Preparation, follows the arrangement instructions, and records any substitutions or additional materials used.
- Inventory updated: Once preparation is complete, the reserved flowers are deducted from physical stock. Any unused flowers return to available inventory.
- Ready for delivery/pickup: Staff mark the order Ready, triggering a customer notification.
- Order completed: Delivery staff confirm delivery, or shop staff confirm pickup. The customer receives a completion notification.
- For cancellations: Reserved stock should be released if preparation hasn't started. Once flowers have been used, staff should record what can be reused and what must be written off.

My priority: I'd like everyone to see the order's current status and accurate inventory levels, especially when we're handling 100+ orders during holidays! 🌷

29. Neil

If an item becomes unavailable before or after checkout, how should substitutions or refunds work, and when must the customer approve a change?

30. Lily

Hi Neil! 🌸 Flower availability can change quickly, so I'd like a fair and transparent substitution process.

- Before checkout: Unavailable flowers should be clearly marked, with similar arrangements or alternative flowers suggested. Customers shouldn't be able to order out-of-stock items.
- After checkout: If a flower becomes unavailable, staff should be alerted immediately and offer a similar flower of equal or greater value at no extra cost.
- Customer approval: Customers must approve changes involving flower types, colors, arrangement style, or additional charges. Minor greenery substitutions can be made without approval.
- Notifications: Customers should receive an email or SMS showing the proposed substitution, any price difference, and a simple way to approve or reject it.
- No response: If customers don't respond before the preparation deadline, staff should contact them directly rather than make significant changes without permission.
- Refunds: If no suitable alternative is accepted, customers should receive a full refund for the unavailable arrangement or item.

My priority: The website must update inventory correctly, release unused flowers, and record all substitutions and customer approvals. I never want customers disappointed by receiving flowers they didn't agree to! 🌷

31. Neil

What confirmations and order-status updates should customers receive, through which channels, and how should they check progress themselves?

32. Lily

Hi Neil! 🌸 I'd like customers to receive clear, timely updates without being overwhelmed by messages.

- Order confirmation: Immediate email with order number, payment receipt, arrangement details, delivery or pickup date, and personal message.
- Order preparation: Email or optional SMS when our florists begin preparing the arrangement.
- Ready for pickup: Email or SMS confirming the flowers are ready, including collection instructions.
- Out for delivery: SMS or email with an estimated delivery window.
- Order completed: Confirmation when flowers are delivered or collected, followed by optional flower care tips.
- Problems or changes: Immediate notifications about delays, substitutions, failed deliveries, or refunds requiring attention.
- Order tracking: Every confirmation email should include a secure tracking link so customers can check their order status without creating an account. Registered customers could eventually view their full order history.

My priority: Email notifications should be automatic, SMS should be optional, and staff should be able to see which notifications were sent. I'd also like customers to receive important updates even if they opt out of promotional messages! 🌷

33. Neil

Should customers be able to check out as guests, create accounts, or both? What should an account let them do that a guest cannot?

34. Lily

Hi Neil! 🌸 I'd definitely like to offer both guest checkout and customer accounts, giving customers the flexibility to choose.

- Guest checkout: Customers can browse, order, pay securely, schedule deliveries, and track orders without registering.
- Customer accounts: Customers can save delivery addresses, recipient details, and preferences for faster future purchases.
- Order history: Registered customers can view previous orders, track current deliveries, and easily reorder favorite arrangements.
- Loyalty rewards: Account holders can earn points, redeem discounts, and access exclusive promotions.
- Personalization: Customers can save favorite arrangements, important dates like birthdays and anniversaries, and choose whether to receive reminders.
- Account security: Customers should be able to update their details, reset passwords, and manage communication preferences.

My priority: Guest checkout is essential for launch, while full customer accounts and loyalty rewards can be introduced in Version 2. I never want customers forced to create an account just to buy flowers! 🌷

35. Neil

Do you offer promotions or a rewards program, and what rules should the site apply when a customer tries to use an expired, invalid, or incompatible offer?

36. Lily

Hi Neil! 🌸 Yes! I offer seasonal promotions and a loyalty rewards program to thank my regular customers.

- Promotions: Discount codes, seasonal specials, referral bonuses, and occasional free delivery offers.
- Loyalty rewards: Customers earn 1 point per $1 spent, with 100 points redeemable for a $5 discount.
- Expired or invalid codes: The website should display a friendly explanation and allow customers to enter another code without losing their cart.
- Incompatible offers: Only one promotional code should apply per order. Loyalty points cannot be combined with other discounts.
- Restrictions: Some offers may exclude wedding packages, custom event orders, and peak holidays such as Valentine's Day.
- Refunds: Points earned from refunded purchases should be reversed, while redeemed points should be returned when appropriate.

My priority: Customers should see their savings clearly before payment, and discounts should never apply incorrectly. I'd like to manage promotions myself, with basic discount codes available at launch and automated loyalty rewards introduced in Version 2! 🌷

37. Neil

What tools currently handle your website, payments, orders, inventory, and customer records, and which need to connect to the new site or remain in use?

38. Lily

Hi Neil! 🌸 At the moment, Lily's Florist uses fairly basic tools, and I'd love to bring everything together!

- Website: A simple website with a contact form, gallery, and business information, but no proper online store.
- Payments: A Square card terminal for in-store payments. I'd prefer to keep Square and connect it to online payments if possible.
- Orders: Phone calls, handwritten notes, and a basic online form. I transfer orders into spreadsheets manually.
- Inventory: Excel spreadsheets and manual stock counts, which often become inaccurate during busy periods.
- Customer records: Basic spreadsheets containing contact information and previous purchases.
- Communication: Gmail for customer inquiries, confirmations, and event bookings.

My priority: I'd like the new website to centralize orders, inventory, and customer information while keeping Square and Gmail. We should also transfer existing customer records and product information without losing important details.

Ultimately, I want to eliminate duplicate data entry and give my staff one reliable place to manage daily operations! 🌷

39. Neil

What pages, product details, photos, and shop policies should appear on the site, and who will provide and keep that information up to date?

40. Lily

Hi Neil! 🌸 I'd love a beautiful, welcoming website that reflects the personality of Lily's Florist and is easy for my team to maintain!

- Essential pages: Home, Shop, Gallery, About Us, Contact, Delivery & FAQs, and Order Tracking. I'd add a Blog and Customer Accounts in Version 2.
- Product details: Each arrangement should show its name, description, price, size options, flower types, availability, care instructions, and delivery options.
- Photos: I'll provide high-quality pictures of our bouquets, seasonal arrangements, wedding flowers, shop interior, and behind-the-scenes work.
- Shop policies: Delivery areas and fees, pickup arrangements, cancellation deadlines, refunds, substitutions, privacy, and payment security.
- Content management: I'll manage prices, promotions, policies, and blog content. My florists should be able to update product photos, descriptions, and flower availability.
- Updates: Inventory should update automatically as orders are processed. We'll review product listings weekly and refresh seasonal content before major holidays.

My priority: I want my staff and me to make everyday website updates ourselves without needing a developer, while keeping all product information accurate and attractive! 🌷

41. Neil

What needs should we accommodate for customers using phones, assistive technology, or languages other than English?

42. Lily

Hi Neil! 🌸 I want Lily's Florist to be welcoming and easy to use for everyone, regardless of their device, abilities, or preferred language.

- Mobile-friendly design: Customers should easily browse flowers, place orders, and make payments using smartphones or tablets.
- Accessibility: Large, readable text, good color contrast, clear buttons, and simple navigation for customers with visual or physical difficulties.
- Assistive technology: The website should support screen readers, keyboard navigation, descriptive image text, and accessible checkout forms.
- Language options: English should be the default, with Spanish available at launch. I'd like to add more languages as customer demand grows.
- AI customer support: Our AI assistant should understand English and Spanish and provide helpful responses in either language.
- Easy checkout: Clear instructions, understandable error messages, and minimal unnecessary steps, especially for older customers.

My priority: Customers should be able to complete an order independently and confidently. I'd also like important information, including prices, delivery policies, and payment instructions, to be equally clear in both languages! 🌷

43. Neil

Which customer questions or tasks should the AI assistant handle around the clock? Could you give an example of a conversation that ends successfully?

44. Lily

Hi Neil! 🌸 I'd love an AI assistant that provides friendly, helpful customer service 24/7, even when my shop is closed!

The AI assistant should handle:

- General questions: Opening hours, shop location, delivery areas, fees, and policies.
- Flower recommendations: Suggest arrangements based on occasion, budget, colors, and preferences.
- Product availability: Check current flower stock, prices, and seasonal specials.
- Order assistance: Help customers place orders, track deliveries, and understand pickup options.
- Flower care: Explain how to keep arrangements fresh and beautiful.
- Language support: Assist customers in English and Spanish.

Example of a successful conversation:

- Customer: I need birthday flowers delivered tomorrow for under $60.
- AI: I'd recommend our $55 seasonal mixed bouquet! Would you like to check delivery availability?
- Customer: Yes, tomorrow afternoon.
- AI: Afternoon delivery is available for your address. The delivery fee is $8, bringing the total to $63 before tax. Shall I show you the checkout page?
- Customer: Yes, please!
- AI: Here's your bouquet ready for checkout. You can add a birthday message and complete your payment securely.

My priority: I'd like AI to resolve at least 70% of routine questions, but always offer human assistance for complaints, refunds, unusual requests, or anything it cannot confidently answer. 🌷

45. Neil

What information should the AI use for its answers, who will keep it current, and does it need access to live inventory or order information?

46. Lily

Hi Neil! 🌸 I'd like our AI assistant to provide accurate, up-to-date information that customers can trust.

- Information sources: Our product catalog, prices, flower care guides, opening hours, delivery policies, FAQs, and seasonal promotions.
- Live inventory: Yes! The AI should check current flower availability before recommending arrangements, especially during busy holidays.
- Order information: Customers should be able to check their order status after securely verifying their identity. The AI must never reveal another customer's details.
- Who maintains it: I'll manage prices, policies, promotions, and FAQs, while my florists update flower availability and product information.
- Automatic updates: Inventory and order statuses should update whenever staff process orders, receive stock, or complete deliveries.
- Accuracy and safety: If information is missing or uncertain, the AI should explain this and refer the customer to my staff rather than guess.

My priority: I want the AI to use the same reliable information my staff see, without independently changing orders, issuing refunds, or promising unavailable flowers! 🌷

47. Neil

If the AI cannot answer, gives an incorrect answer, or a customer requests a person, how should it respond and arrange follow-up, especially outside business hours?

48. Lily

Hi Neil! 🌸 I want customers to feel supported, even when our AI assistant cannot resolve their concerns.

- Unable to answer: The AI should politely explain its limitations and offer to connect the customer with a staff member.
- Incorrect answers: It should apologize, correct the information using verified shop details, and alert staff if the mistake affected an order.
- Human assistance: Customers should always have the option to request a person, without repeatedly explaining their problem.
- During business hours: Staff should receive an immediate notification and take over the conversation when available.
- Outside business hours: The AI should collect the customer's name, contact details, issue, and order number if relevant, then create a follow-up request for the next business day.
- Urgent problems: Delivery failures, incorrect orders, and payment issues should receive priority attention.

My priority: Customers should receive a reference number and confirmation of their request, with a clear expectation of when staff will respond. I'd like staff to see the conversation history so customers don't have to repeat themselves. 🌷

49. Neil

What customer information may the AI access or collect, how should it verify someone before discussing an order, and which actions must require staff approval?

50. Lily

Hi Neil! 🌸 Protecting my customers' personal information is extremely important to me!

- Information collected: The AI may ask for names, email addresses, phone numbers, delivery preferences, and order numbers when needed.
- Information accessed: It should only access relevant product details, order statuses, and delivery information necessary to answer a customer's question.
- Identity verification: Before discussing a specific order, customers should sign into their account or verify themselves using a one-time code sent to the email or phone number associated with the order.
- Privacy: The AI must never reveal payment card details, passwords, or another customer's information. It should only collect essential details and avoid retaining unnecessary personal information.
- Staff approval required: Refunds, cancellations, significant order changes, flower substitutions, price adjustments, and changes to delivery addresses after dispatch.
- Customer consent: The AI should clearly explain when information is collected and when a conversation is passed to staff.

My priority: The AI should help customers confidently without compromising their privacy. I'd also like staff to review AI conversations when resolving problems, with access restricted to authorized employees. 🌷

51. Neil

What launch date, budget, and operational constraints should guide the project, including expected normal and peak order volumes?

52. Lily

Hi Neil! 🌸 I'd love to have our new website ready well before Valentine's Day, our busiest season!

- Launch date: Ideally, January 15, 2027, allowing time for testing and staff training before Valentine's Day.
- Budget: I'm comfortable investing $12,000–$18,000 initially, with approximately $300–$500 monthly for hosting, maintenance, and AI services.
- Normal order volume: We currently receive 20–30 orders daily, but I'd like to grow to 40–50 daily orders.
- Peak order volume: Valentine's Day, Mother's Day, and major holidays can bring 100+ orders daily. The website should comfortably support 150–200 orders per day.
- Operational constraints: We're a small team, so the system must be simple to use, minimize manual tasks, and integrate with Square and Gmail.
- Other requirements: Existing customer records must be preserved, staff need training, and the website must work reliably on mobile devices.

My priority: I'd rather launch a reliable website with essential features on time than rush advanced features. We cannot afford major disruptions to orders or deliveries during our busiest holidays! 🌷

Appendix B: Project Checklist

Summary

Complete the project individually, starting with the Lily interview. Follow the steps in order, checking each off when its stated result exists.

The checklist targets the rubric's highest-score criteria while keeping the work simple. It covers planning and design; building a working website is not required.

Source: School project rubric.

1. Interview Lily

☑ **Prepare interview questions.** Cover current frustrations, business goals, customers, ordering, inventory, and round-the-clock AI support.

☑ **Open the school-provided Lily chat.** Introduce yourself and explain what you want to learn.

☑ **Understand the current process.** Ask how orders, inventory, and customer support work today and where problems occur.

☑ **Clarify the desired website.** Ask what customers and staff should accomplish and which outcomes matter most.

☑ **Clarify AI support.** Ask what it should handle, what information it needs, and when human help is necessary.

☑ **Explore problems and exceptions.** Discuss a successful order, then situations such as unavailable products, failed checkout, or unanswered support questions.

☑ **Confirm your understanding.** Summarize Lily's needs back to her and resolve gaps or contradictions.

☑ **Save the evidence.** Keep the transcript and create the share link required for submission.

2. Write the user stories

☑ **Summarize the interview.** Record confirmed needs, user types, and any remaining assumptions separately.

☑ **Write at least 15 distinct user stories.** Use: "As a [role], I want [need], so that [outcome]."

☑ **Check story coverage.** Include multiple user types and AI-enabled customer support; ensure ordering and inventory needs from the interview are represented.

☑ **Number and verify the stories.** Label them `US-01` through `US-18` and record the supporting interview answer for each. Ask Lily follow-up questions if needed.

3. Map one user journey

☑ **Choose one goal from the stories.** Select a concrete task that will also work well as a wireframe sequence.

☑ **Define the persona and scenario.** State who the user is, what prompted the task, and what success means. List the related story IDs.

☑ **Map the successful path.** For each step, record the user's action, the system's response, and the next step.

☑ **Add relevant alternate paths.** Show likely problems, dead ends, and how the user recovers or gets help.

4. Define the requirements

☑ **Draft at least 20 requirements from the stories.** Include website functionality and AI-enabled customer support.

☑ **Number and categorize them.** Use `FR-01...` for functional requirements and `NFR-01...` for nonfunctional requirements. Include both categories.

☑ **Make each requirement verifiable.** Replace vague wording with observable behavior or measurable constraints. Confirm important unspecified targets with Lily rather than inventing them.

☑ **Connect requirements to stories.** Put the supporting story IDs beside every requirement.

☑ **Review the complete set.** Check for ambiguity, contradictions, duplicates, and missing customer needs.

5. Create the Figma wireframes

☑ **List the screens and states needed for the journey.** Include the main path and relevant problem or recovery states.

☑ **Draw rough wireframes in Figma.** Use simple boxes, labels, and controls; prioritize a clear flow.

☑ **Annotate the wireframes.** Explain design choices, what happens when controls are used, and which requirement IDs the elements address.

☑ **Walk through the journey using the wireframes.** Fix missing steps, confusing interactions, and inconsistencies with the requirements.

6. Assemble and submit

☑ **Assemble one document.** Include the interview share link, user stories, user journey, requirements, and readable wireframe images with annotations.

☑ **Check it against the rubric.** Verify the counts, multiple user perspectives, AI support coverage, both requirement categories, and traceability.

☑ **Export and inspect one PDF of at least four pages.** Check spelling, layout, image legibility, and that the interview link is accessible.

☑ **Submit through the school dashboard.** Retain a copy of the submitted PDF.

Defaults

The original targets were at least 15 stories and 20 requirements. This writeup contains 18 stories, 40 functional requirements, and 5 nonfunctional requirements, with five wireframes covering one journey and its out-of-stock branch.

The working checklist is available in project_checklist.md.

Appendix C: Wireframe Design Conversation

The following transcript records the Figma Make design and export process. Build and export statements are reproduced from that conversation.

Wireframe Turn 1: User

Aim to deliver your final answer or completed task within 60 seconds. Keep the scope tight and answers concise. If the task needs longer, give me a useful result and clearly state what remains by 60 seconds. Never claim unfinished work is complete.

I have a project that I need wireframes for. Specifically, we are creating a mockup of a user journey for a hypothetical customer called, "Lily's Flower Shop". See the attached markdown document, read through it for context, and build a set of wireframes based on the section titled `## User Journey: Finding Flowers with AI Assistance`

Wireframe Turn 2: Assistant

I'll retrieve the brief, inspect the existing app structure and styling, then implement the requested user-journey wireframes as a tightly scoped flow.

Wireframe Turn 3: Assistant

The brief defines four core states plus an out-of-stock branch; I'm translating those into a responsive storyboard with catalog, search, AI inventory check, and cart/checkout wireframes, including a toggleable alternative-color scenario.

Wireframe Turn 4: Assistant

The full four-step flow is now implemented, including the AI success/fallback branch and synchronized product/cart state; I'm running the project build to catch any TypeScript or styling integration issues.

Wireframe Turn 5: Assistant

Implemented the responsive wireframe journey for **Lily's Flower Shop**:

- Shop catalog
- Rose search results with pricing and availability
- AI inventory assistant
- Toggleable in-stock/out-of-stock scenarios
- Alternative flower recommendation
- Product selection and cart checkout handoff
- Desktop and mobile-responsive layouts

Verification: `pnpm build` completed successfully.

Wireframe Turn 6: User

Ok, this needs to be a mobile-first design, though

Wireframe Turn 7: Assistant

I'll recast the storyboard around phone-sized screens: compact mobile navigation, two-column product browsing, full-width AI chat, and a bottom-sheet cart, while retaining a useful desktop overview of all four mobile states.

Wireframe Turn 8: Assistant

The flow now uses true phone-width canvases and mobile interaction patterns rather than scaled desktop pages; I'm checking the revised implementation for build errors.

Wireframe Turn 9: Assistant

Updated the wireframes to a mobile-first design:

- Phone-sized canvases for all four journey steps
- Mobile navigation and status bar
- Two-column product grids
- Full-screen AI chat experience
- Mobile bottom-sheet cart
- Single-column flow on small screens
- Two-column phone overview on larger screens

`pnpm build` passes successfully.

Wireframe Turn 10: User

Can you add annotations that describe each of the core elements described in user stories, e.g. nav buttons, dropdowns, drawer buttons, etc.?

Wireframe Turn 11: Assistant

I'll map the referenced user stories to their specific controls, then add numbered callouts and a concise annotation key beside each mobile wireframe.

Wireframe Turn 12: Assistant

The source requirements confirm the key behaviors: top inventory navigation, category access, availability states, AI launch/inventory response, plus-control add-to-cart, cart drawer, and checkout handoff. I'm attaching those directly to visible controls with story-linked callouts.

Wireframe Turn 13: Assistant

The annotations are now attached to the actual UI controls and cross-referenced to the relevant stories; I'm running the build once to verify the added components and conditional markers.

Wireframe Turn 14: Assistant

Added numbered, story-linked annotations for:

- Top Shop and Cart navigation
- Category dropdown and tabs

- Search field and inventory results
- AI launcher, conversation drawer, response, and composer
- Add-to-cart control and quantity confirmation
- Mobile cart drawer
- Checkout button and handoff behavior

Each mobile wireframe now includes a matching annotation key referencing `FT-01` or its associated user story.

Verification: `pnpm build` passes.

Wireframe Turn 15: User

Can you create a set of PNG images from this mockup that can be used in a presentation document?

Wireframe Turn 16: Assistant

I'll capture presentation-ready PNGs of the complete storyboard and each annotated mobile step, then place the exports in a clearly named project folder.

Wireframe Turn 17: Assistant

The supervised preview is available on port 3000, so I'll export one high-resolution overview plus individual PNGs for all four steps and the AI out-of-stock branch.

Wireframe Turn 18: Assistant

The renderer is producing the six high-resolution exports now; I'm waiting for file validation so I can confirm exact dimensions and filenames.

Wireframe Turn 19: Assistant

The first capture attempt stalled on an unavailable external font request; I've isolated that issue and am rerunning with local fallbacks so the PNG export itself is not blocked.

Wireframe Turn 20: Assistant

The local-font fallback resolved the renderer stall and the test PNG validated at 1040×2160; I'm generating the final named set at 2× presentation resolution.

Wireframe Turn 21: Assistant

Five individual exports are complete; the oversized browser overview was unreliable, so I'm assembling a clean 2×2 presentation board from the validated step images instead.

Wireframe Turn 22: Assistant

The visual check caught two annotation markers overriding positioned controls; I'm correcting the AI launcher and cart drawer positioning, then regenerating the affected PNGs and overview.

Wireframe Turn 23: Assistant

Created six presentation-ready PNGs in:

`exports/lilys-flower-shop-wireframes/`

Files:

- `00-complete-journey.png` — 2×2 overview board
- `01-visit-shop.png`
- `02-search-roses.png`
- `03-ai-in-stock.png`
- `03b-ai-out-of-stock.png`
- `04-cart-checkout.png`

Individual slides are **1040 × 2160 px**. The overview is **1344 × 2636 px**.

Wireframe Turn 24: User

Can you quickly export our conversation (just the chat text) in markdown format?

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